

Monte Carlo Tree Search for the Enzyme Mutation Game: Discovering Epistatic Pathways in Protein Sequence Space

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Abstract

Predicting the effects of multi-point mutations in enzymes remains a central challenge in protein engineering due to epistasis and the vastness of sequence space. We formalize enzyme mutagenesis as a sequential decision-making game—the *Enzyme Mutation Game*—and propose a novel framework combining **Monte Carlo Tree Search (MCTS)** with **Deep Graph Translation Networks (DGTN)** to discover high-fitness mutation pathways. Our method treats each amino acid substitution as an action and models evolutionary trajectories through a learned fitness landscape. The DGTN encodes sequences into a structured latent graph, enabling interpretable reasoning over residue-residue interactions, while MCTS performs strategic planning to navigate epistatic dependencies. We introduce a mechanism to reconstruct and visualize the hidden fitness landscape using active sampling and Gaussian process regression. Evaluated on synthetic landscapes with known epistatic pairs and extended to real enzyme families via ESM-2 embeddings, our approach outperforms random screening and gradient-free baselines in finding global optima. This work establishes a new paradigm for *explainable AI in directed evolution*, where search traces reveal not only optimal sequences but also the mechanistic logic behind them.

Keywords: Protein engineering, $\Delta\Delta G$ prediction, Graph neural networks, Transformers, Multi-modal learning, Interpretable AI, Rational protein design

1 Introduction

Protein thermodynamic stability, quantified by the change in Gibbs free energy upon mutation ($\Delta\Delta G = G_{\text{mutant}} - G_{\text{wild-type}}$), is a fundamental property that governs protein folding, function, and evolutionary fitness ?. Accurate prediction of mutation effects enables rational protein engineering for diverse applications including therapeutic antibody development ?, industrial enzyme optimization (?), and understanding disease-causing mutations ?.

Traditional computational methods employ physics-based energy functions (FoldX ?, Rosetta ?), achieving moderate accuracy (Pearson $\rho \approx 0.5$) but requiring extensive computational resources. Directed evolution has revolutionized enzyme engineering by mimicking natural selection in the lab ?. However, its efficiency is limited by combinatorial explosion: even a single-site saturation scan across 20 amino acids at 10 positions yields 20^{10} variants—far beyond experimental throughput. The root cause of this complexity is *epistasis*: non-additive interactions between mutations ?. A mutation that is neutral alone may become beneficial or deleterious depending on genetic background. This creates rugged, multi-modal fitness landscapes that resist greedy optimization.

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Recent machine learning approaches leverage deep neural networks, improving accuracy to $\rho \approx 0.75 - 0.80$???. Recent advances in deep learning offer new tools. Protein language models (pLMs) like ESM-2 ? capture evolutionary constraints from sequence data. However, they are primarily used for point predictions, not for *planning* mutation pathways. However, these methods face three critical limitations:

Challenge 1: Interpretability Gap. Black-box models provide predictions without mechanistic explanations, hindering adoption by experimental researchers who need to understand *why* a mutation is predicted to stabilize or destabilize a protein.

Challenge 2: Modal Integration. Proteins exhibit multi-scale organization—sequence determines structure, which determines function. Existing methods process sequence and structure independently, missing critical cross-modal dependencies.

Challenge 3: Strategic Reasoning. Protein design requires systematic exploration of mutation space with multiple competing objectives (stability, solubility, activity). Current approaches evaluate mutations individually without strategic search capabilities.

Our Contributions

In this paper, we reframe enzyme engineering as a *game*, inspired by AlphaGo ?. Each move is a point mutation; the goal is to maximize long-term fitness. We present **EnzyMCTS**, a framework that combines:

Monte Carlo Tree Search (MCTS) for efficient exploration of sequence space,
Deep Graph Translation Networks (DGTN) to model structural and functional constraints,
Fitness landscape reconstruction for visualization and scientific insight.

We present an **Interpretable Multi-Modal Reasoning Framework** that addresses these challenges through:

1. **Hybrid GNN-Transformer Architecture (§??):** Novel cross-modal fusion integrating geometric graph neural networks for 3D structure with transformers for sequence, using mutation-aware attention mechanisms.
2. **Strategic Search Framework (§??):** Five complementary strategies (greedy, beam search, Monte Carlo, ensemble, constraint-based) with rigorous mathematical formulation and convergence analysis.
3. **Chain-of-Thought Reasoning (§??):** Automated generation of human-interpretable explanations grounded in structural biology principles, with confidence calibration and molecular rationale.
4. **Comprehensive Evaluation (§??):** Achieves state-of-the-art performance ($\rho = 0.85$) on ProTherm with superior interpretability (human evaluation: 4.2/5.0), demonstrated through three real-world case studies.
5. Formalization of the *Enzyme Mutation Game* as a Markov Decision Process (MDP).
6. Integration of DGTN with MCTS for interpretable, structure-aware planning.
7. A method to infer and visualize the hidden fitness landscape using active querying.
8. Demonstration of superior performance in discovering epistatic optima.

The rest of this paper is organized as follows: Section 2 reviews related work. Section 3 details the prediction model architecture. Section 4 presents the reasoning framework with mathematical formulation. Section 5 describes experiments and results. Section 6 discusses findings and limitations. Section 7 concludes with future directions.

2 Related Work

The progression of computational methods for predicting the change in stability upon mutation ($\Delta\Delta G$) has evolved through three distinct eras, each marked by a trade-off between speed, accuracy, and interpretability. The initial phase focused on **physics-based approaches** such as FoldX [1] and Rosetta [2], which employed molecular mechanics force fields and rigorous energy minimization techniques. While these methods offered mechanistic interpretability by directly modeling physical interactions, they were computationally expensive, often requiring minutes per mutation, and yielded only moderate accuracy. The subsequent development of **statistical machine learning (ML) techniques** utilized hand-crafted features coupled with classical algorithms, exemplified by I-Mutant2.0 [3] and the work of Cheng et al. [4]. This statistical approach increased speed and incrementally improved accuracy to the range of $\rho \approx 0.6 - 0.7$.

The field achieved its current state-of-the-art accuracy with the advent of **deep learning architectures**. This era bifurcated into two main strategies: **sequence-based deep learning** and **structure-based deep learning**. Sequence-based methods, notably Protein Language Models (PLMs) such as those developed by Meier et al. [5] and Rives et al. [6], achieved the highest predictive performance, with correlations reaching $\rho \approx 0.75 - 0.78$ by leveraging unsupervised pre-training on millions of evolutionary sequences. Conversely, **structure-based deep learning** utilized Graph Neural Networks (GNNs) and 3D Convolutional Neural Networks (CNNs), including DeepDDG [7], Li et al.’s work [8], and Zhou et al.’s model [9], to explicitly model the protein’s tertiary structure. While these models successfully captured local physical interactions, they typically lagged slightly in accuracy ($\rho \approx 0.70 - 0.73$) due to their reduced ability to incorporate large-scale evolutionary context. Initial **hybrid approaches** attempted to combine these modalities via simple concatenation or late fusion, but they lacked a principled mechanism for truly synergistic cross-modal interaction.

Despite achieving high numerical accuracy, current deep learning paradigms present critical limitations for rational, high-stakes therapeutic design. First, the models primarily rely on post-hoc interpretation methods—such as attention visualization [10], saliency maps [11], and concept-based explanations [12]—which provide suggestive input correlations rather than integrated, human-understandable mechanistic reasoning. This opacity fundamentally undermines the trust required for clinical translation. Second, existing design frameworks, including those based on genetic algorithms [13], Bayesian optimization [14], and reinforcement learning [15], treat the design process as a heuristic search through a vast sequence space. Critically, these methods fail to integrate the prediction and search steps with explicit, verifiable mechanistic reasoning, thus providing design choices without scientific justification and lacking the strategic search capabilities necessary for truly efficient and accountable protein engineering.

3 Methodology

3.1 Problem Formulation

Input:

- Protein sequence $\mathbf{s} = (s_1, \dots, s_L)$ where $s_i \in \mathcal{A}$ (20 amino acids)
- 3D structure graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V}$ represents residues with coordinates $\{\mathbf{r}_i\}$
- Mutation $m = (p, s_p^{\text{wt}}, s_p^{\text{mut}})$: position, wild-type, mutant

Output:

- $\Delta\Delta G$ prediction with confidence $c \in [0, 1]$
- Reasoning chain $\mathcal{R} = \{r_1, \dots, r_k\}$ explaining the prediction
- Molecular rationale connecting prediction to structural/sequence features

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